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$$f(x) = x + 2, g(x) = -3x \text{ en } h(x) = \frac{2-x}{3}$$

$$\text{Eerst: } f(x) = h(x)$$

$$x + 2 = \frac{2-x}{3}$$

$$x = -1$$

$$f(-1) = 1$$

$$\Rightarrow (-1, 1)$$

$$\text{Ten tweede: } f(x) = g(x)$$

$$x + 2 = -3x$$

$$4x = -2$$

$$x = -\frac{1}{2}$$

$$f\left(-\frac{1}{2}\right) = \frac{3}{2}$$

$$\Rightarrow \left(-\frac{1}{2}, \frac{3}{2}\right)$$

$$\text{Ten laatste: } g(x) = h(x)$$

$$-3x = \frac{2-x}{3}$$

$$-8x = 2$$

$$x = -\frac{1}{4}$$

$$g\left(-\frac{1}{4}\right) = \frac{3}{4}$$

$$\Rightarrow \left(-\frac{1}{4}, \frac{3}{4}\right)$$

$$\text{Wij hebben twee intervallen } \left[-1, -\frac{1}{2}\right] \text{ en } \left[-\frac{1}{2}, -\frac{1}{4}\right]$$

Vul waarden tussen deze intervallen in en zie welke boven- en onderaan is:

$$\text{Eerste gebied: } x \in \left[-1, -\frac{1}{2}\right]$$

$$v_1(x) = f(x) - h(x) = x + 2 - \frac{2-x}{3} = \frac{4x+4}{3}$$

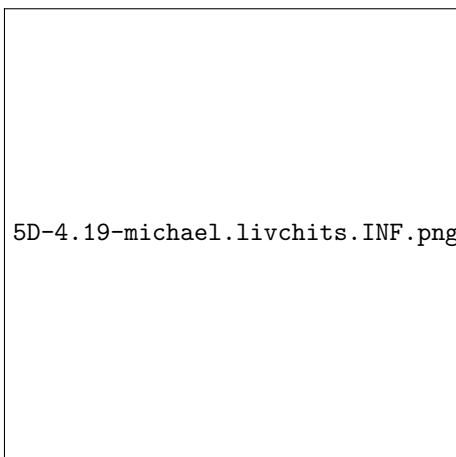
$$\text{Tweede gebied: } x \in \left[-\frac{1}{2}, -\frac{1}{4}\right]$$

$$v_2(x) = g(x) - h(x) = -3x - \frac{2-x}{3} = \frac{-8x-2}{3}$$

$$I = \int_{-1}^{-\frac{1}{2}} \left(\frac{4x+4}{3} \right) dx + \int_{-\frac{1}{2}}^{-\frac{1}{4}} \left(\frac{-8x-2}{3} \right) dx$$

$$= \left[\frac{2x^2+4x}{3} \right]_{-1}^{-\frac{1}{2}} + \left[\frac{-4x^2-2x}{3} \right]_{-\frac{1}{2}}^{-\frac{1}{4}}$$

$$= \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$$



References